A

PROJECT REPORT

ON

"M-Training Toolkit on Polio / Routine Immunization for CMCs in CGPP."

Submitted in the partial fulfillment of the requirement for the award of degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING

Submitted To:

Ms. Kavita Khanna
HOD, CSE

Submitted By:

Anubha Parashar
4957592
CSE 8^{th} SEM

Under the Guidance of:
Mr. Subhi Quraishi
CTO,Mobility Division

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
PDM COLLEGE OF ENGINEERING, BAHADURGARH
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Mobile 4 Development

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CERTIFICATE

I hereby certify that the Project Work entitled **"M-Training Toolkit on Polio / Routine Immunization for CMCs in CoreGroup Polio Project."**, which is being submitted by Ms. Anubha Parashar (9142) in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering submitted, PDM College of Engineering, Sarai Aurangabad, Bahadurgarh, Haryana , is an authentic record of my own work carried out under the supervision of Mr. Subhi Quraishi.

The matter presented in this thesis work has not been submitted for the award of any other degree of this or any other university.

This is to certify that the above statement made by the candidate is correct and true to the best of my knowledge and belief.

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ZMQ Software Systems INDIA

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ABBREVIATIONS

CGPP	Core Group Polio Project
CORE	Child Survival Collaborations and Resources Group
PCI	Project Concern International
ADRA	Adventist Development & Relief Agency
CRS	Catholic Relief Services
SA	Super Administration
PL	Partners Management Layer
SRC	State Regional Coordinators
DMC	District Management Coordinators
BMC	Block Management Coordinators
CMC	Community Mobilizers Coordinators
PHC	Physical Health Centers
SIA	Supplemental Immunization Activities
RI	Routine Immunization
OPV	Oral Polio Vaccine
SM Net	Social Mobilization Network

NOMENCLATURE

English Symbols	
P	Polio drop given at home
Pb	Polio drop given at booth
XL	Left over house
XS	Family declines vaccination because the child is sick
XV	Child is out of the village/town and is not expected to return before the end of the round
XH	Child is not at home but will return
XR	Refusal by family

DECLARATION BY THE CANDIDATE

I, **Anubha Parashar,** hereby declare that the project work entitled **"M-Training Toolkit on Polio / Routine Immunization for CMCs in CoreGroup Polio Project"** is an authenticated work carried out by me at **ZMQ Software Systems** for the partial fulfillment of the award of the degree of Bachelor of Technology in Computer Science & Engineering submitted at, PDM College of Engineering, Bhadurgarh, Jhajjar, Haryana. This work has not been submitted for similar purpose anywhere else.

Date: 21-05-13
Place: Sector-7, IMT, Manesar, Gurgaon.

Anubha Parashar
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B.Tech CSE 8th Sem

ACKNOWLEDGEMENT

Many individuals have contributed to this project report with their advice, interest, time and support. They are the wind beneath my wings. I thank God for making everything possible.

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I would specially like to thank my worthy guide **Mr. Subhi Quraishi**, ZMQ who supervised my project. His technical advice, ideas and constructive criticism contributed to the success of this report. He suggested me many ideas and solved my puzzles when I was in need. His motivation and help has been of great inspiration to me.

I would also like to thanks my respected HOD (CSE) and faculty members of CSE department for providing me the opportunities, support and the necessary help to complete this project work.

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ABSTRACT

I'll be working with ZMQ in the Mobile for Development group of the company during my industrial training. In the M4D group, I will be part of its Core Group-ZMQ Polio project. The Core Polio project is developing multi-layered technology tools for different tiers of health providers in 19 high risk districts of U.P. and 3 districts of West Bengal for efficient management, demand generation and creating awareness around Polio and need of other routine immunization in these regions. Variety of technology tools, systems and platforms are being developed as part of the Polio/ Routine Immunization Management System.

I will be working with the team developing mobile toolkit for last level health workers called the Community Health Mobilizers (CMCs). The toolkit developed for CMCs have many components and I will be developing the **mobile phone based training and community engagement module for CMC**. The mobile training toolkit will be a miniature client/server m-learning toolkit for CMCs. The objective of the toolkit is to delivery quick and compact training modules to CMCs on various immunization issues under the Universal Immunization programs of the Government of India. It will also have community training modules on issues of concern for community members. The toolkit will be developed with iconic language for clients having low-level of literacy.

During the course of the training, I will also be developing **a basic game** for mobile devices focusing on the needs of semi-literate audiences in the developing world. I will be designing and developing solutions for low-end handsets and feature phones catering to poor and resource constraint populations. I will be working on Java (J2ME Platform) with MS SQL as backend.

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CHAPTER 1

INTRODUCTION

The primary strategy to interrupt transmission of wild poliovirus in India is to improve supplemental immunization activities and routine immunization coverage in priority districts with a focus on 107 high-risk blocks of western Uttar Pradesh and central Bihar. Villages or urban areas with a history of wild poliovirus transmission, or hard-to-reach or resistant populations are categorized as high-risk areas within blocks. The Social Mobilization Network (SM Net) is formed in Uttar Pradesh to support polio eradication efforts through improved planning, implementation and monitoring of social mobilization activities in those high-risk areas.

1.1 BACKGROUND

Almost all wild polio cases in India are from high-risk districts in western Uttar Pradesh and central Bihar. The most recent primary strategy to interrupt transmission of wild poliovirus in India is to improve supplemental immunization activities (SIAs or mass campaigns) and routine immunization coverage in 107 high-risk blocks of western Uttar Pradesh and central Bihar. It is widely acknowledged that grass roots social mobilization efforts are needed to reach underserved populations during SIAs and to combat rumors against polio vaccination. The CORE Group is a US-based organization made up of health professionals, working for a variety of non-governmental organizations, who collaborate on international health and development programs. In 1999, the United States Agency for International Development (USAID) launched the CORE Group Polio Project (now known as CGPP) in six countries, including India. The CGPP harnesses and synchronizes the efforts of a coalition of US-based Private Voluntary Organizations (PVOs) and their in-country offices, as well as non-governmental organization (NGOs) partners to support the polio eradication effort by providing both social mobilization and detailed local planning for vaccination services. The CGPP focuses on targeting the most inaccessible populations, whether due to cultural or physical barriers to access. The project reaches these populations by systematic enumeration and tracking of children less than five years, and through highly targeted social mobilization strategies that rely on direct personal communication with families and with informal and formal community leaders.

In India, the CGPP works in ten districts of the state of Uttar Pradesh (UP) through a consortium of the following PVOs: Adventist Development & Relief Agency (ADRA) India, Project Concern International (PCI) and Catholic Relief Services (CRS), as well as their local NGO partners. The CGPP India has a Secretariat providing coordination and technical and managerial support to field staff. The CGPP India has an extensive network of 1,325 Community Mobilization Coordinators (CMCs) who conduct social mobilization activities for behavior change related to polio vaccination. These CMCs are a part of the Social Mobilization Network (SM Net) in India that includes CORE, UNICEF, Rotary, and the Indian Government's and WHO's National Polio Surveillance Project (NPSP).

1.2 PURPOSE

- The CMC reach out to the very poor and they educate and immunize local communities to eradicate polio.
- Reaching the high risk groups and Vulnerable Communities
- Tracking missed children for Immunization
- Building Health system’s capacities to deliver immunization program
- Developing Systems for community demand generation.
- Strengthening National and Regional Immunization Systems
- Instant Data Collection, Mapping & Digital Advisory Service Delivery
- Supported Timely Documentation and Use of Information For Game

1.3 LITERARY SURVEY OF THE SUBJECT

Each of the past three centuries has been dominated by single technology. The 18th century was the time of the great mechanical system accompanying the industrial revolution. The 19th century was the age of the steam engine, while during the 20-21st century; the key technology has been the information gathering, processing, and distribution. Among other development we have seen the installation of the worldwide telephone networks, the invention of the Radio, Television, the birth and high growth of computer industry, and the launching of the communication satellite.

Technology progress from square of micro-to-micro electronics. Due to rapid technology progress, and thesize of the computer are reduces. Mobile devices are more portable. The use of Mobile device is increasing day by day. We are communicating with other place by using Mobile devices are communication satellite. We can also play games at few minutes free time to get some relaxation.

1.4 METHODS

We carried out a secondary data analysis of routine monitoring information collected by the SM Net and the Government of India. These data include information about vaccination outcomes in SM Net areas and non-SM Net areas within the districts where the CORE Group operates. Statistical analysis was used to compare, between SM Net and non-SM Net areas, vaccination outcomes considered sensitive to social mobilization efforts of the SM Net.

1.5 RESULTS

Vaccination outcomes in SM Net areas were as high as or higher than in non-SM Net areas. There was considerable variation in vaccination outcomes between districts.

1.6 INTRODUCTION OF THE ORGANIZATION

ZMQ software system is an Innovative Software Solution Provider specializing in customized eLearning solution, Edutainment system, Gaming solutions, Knowledge Management System, Human Performance solutions and Training Needs Analysis.

ZMQ has expertise in Analysis & Research, Industrial Design, and Development, of any kind of learning solution. The application of our learning solutions include the areas of Simulated Courseware Development & Authoring, Industrial Design, Simulated Laboratories, Health, Environment, Multi-cultural Educational platform, Educational Games, eQuizzing System, Virtual Tourism, Social Learning Solutions, Needs Analysis and Training Management System.

ZMQ has an outstanding reputation and track record. ZMQ has provided effective and affordable tailor-made eLearning solutions to Academic Institutions, eLearning Companies, Corporate and NGOs in almost all the major regions of the world. ZMQ also focuses on developing world-class learning and edutainment solutions for semi-literates, under-privileged, physically handicapped and out-of-school children.

To enhance human performance, ZMQ's solutions are designed instructionally sound using best learning methodologies and human communication theories. The solutions are further blended with high-end graphics, simulations and animation, and converting static content into differentiated experiential content. This not only helps in effective, quicker and easier learning but also increases retention of information and helps to built skills and competence. The performance is monitored and the success is evaluated through a sophisticated management system using latest technologies.

ZMQ has capabilities to deliver solutions on a wide variety of technologies from Internet based community Learning system, Intranet based Closed Net Learning, Stand-alone Learning (CBTs) to compact Learning on Hand-held & mobile Devices.

Our Mission is to build a better world by turning ideas and concepts into innovative and interactive solutions that educate and entertain the mind and at the same time monitor and evaluate the performance.

Our Vision is to deliver highly innovative and cost effective products that demonstrate our care for our customers.

ZMQ's motto is to Work across Culture and Continents to expand Knowledge and distribute Information.

1.6.1 Services offered by ZMQ

ZMQ is a one-stop service provider that offers effective and efficient solutions tailored to the clients business needs. Clients can outsource different services and operations either selectively or cumulatively. ZMQ look for a long-term arrangement with the client as their IT partner to provide its services for various business needs and functions.

The areas that are covered include:

- Software Development
- Consulting Services
- Support Services

1.6.1.1 Software Development

The software development division undertakes on-site development work and offshore development jobs as well as software maintenance task.

If the client wants the development work to be carried out at their site, ZMQ has dedicated teams that can take up such projects. These teams can handle such projects independently or together with the team members of the clients.

ZMQ allows the clients to enjoy all the benefits of a dedicated development facility without the responsibility of setting up and managing one. The development work can be done at its offshore facilities having the entire required infrastructure.

1.6.1.2 Consulting Services

The Consulting Services Division offers a broad range of technical services to the clients to enable them to use IT for the development and growth of their organizations.

1.6.1.3 Support Services

ZMQ also provides a wide range of support services to the clients. The organizations need not employ qualified and skilled personnel or dedicate valuable in-house resources to non-core activities like support and maintenance. Outsourcing the support and maintenance tasks to a competent agency like ZMQ may enable the organization to be completely free from the hassles of recruiting and retaining professionals, and providing them with training and retraining. It has a large pool of engineers and technical staff to endure regular and effective management and support for systems, networks and software. It also takes care of the routing jobs such as backup and restores services and disaster management and recovery.

1.6.1.4 Software Solutions

1. Global Learning Solutions
2. Enterprise Information and Management Solutions
3. Training Needs Analysis (TNA).
4. Database Management Systems
5. Knowledge Management System
6. Mobile Learning, Gaming & Edutainment Systems

1.7 INTRODUCTION OF M-HEALTH SOLUTIONS FOR LOW RESOURCE SETTING

M-Health is a term that means “mobile health,” as in harnessing the power of wireless technology to support healthcare provision, research or education. The infrastructure for M-health is already in place as there are over 4.1 billion cell phones in the world.

Within the larger field of global health, M-health efforts may have the most potential for impact within community-oriented programming, due to the "off site" and sometimes remote nature of public health work designed to reach underserved populations. “M-health community of practice” in response to interest in community-level applications that can increase quality, equity scale and/or sustainability of health efforts.

The goal: measurable impact on the health of underserved women and children in low- and middle-income countries.

1.8 Introduction of Mobile Applications:

Ever since the mobile phones came into existence they were loaded with basic standalone applications like Calendar, Organizer and Clocks and Alarms. This led to multi-functionality of the mobile phone and gradually people started using mobiles not only for making phone calls but also for doing other activities. As the capabilities of devices grew so did the need to expand the opportunities of using mobile phones for other activities. With time mobile phone became an attractive platform for developing both standalone and connected applications. With this came in being a new domain of technology solutions like M–Commerce, m-Learning and M-entertainment. The project take the scope of connected mobile application to another step, that is developing Mobile connected platforms for SM net. The developed app is designed to connect the large number of communities to their health workers in small towns and villages of India for managing vaccination.

The connected Mobile Applications are developed on variety of Mobile platforms like:
1. BREW on CDMA platform
2. Java /J2ME for GSM Technology

1.9 OUTLINE OF THE PROJECT

CGPP's India Secretariat is located in Delhi and consists of a team of 4 consultant advisors, which operates through three CORE Group members (Project Concern International, Adventist Development and Relief Agency India, and Catholic Relief Services) and 11 national NGO partners. The project is implemented via an extensive network of **Community Mobilization Coordinators (CMCs)** who conduct social mobilization activities in high-risk areas to promote acceptance of the oral polio vaccine (OPV).

A three-tier network of CGPP mobilizers effectively sustains the changed behavior of the target audience.

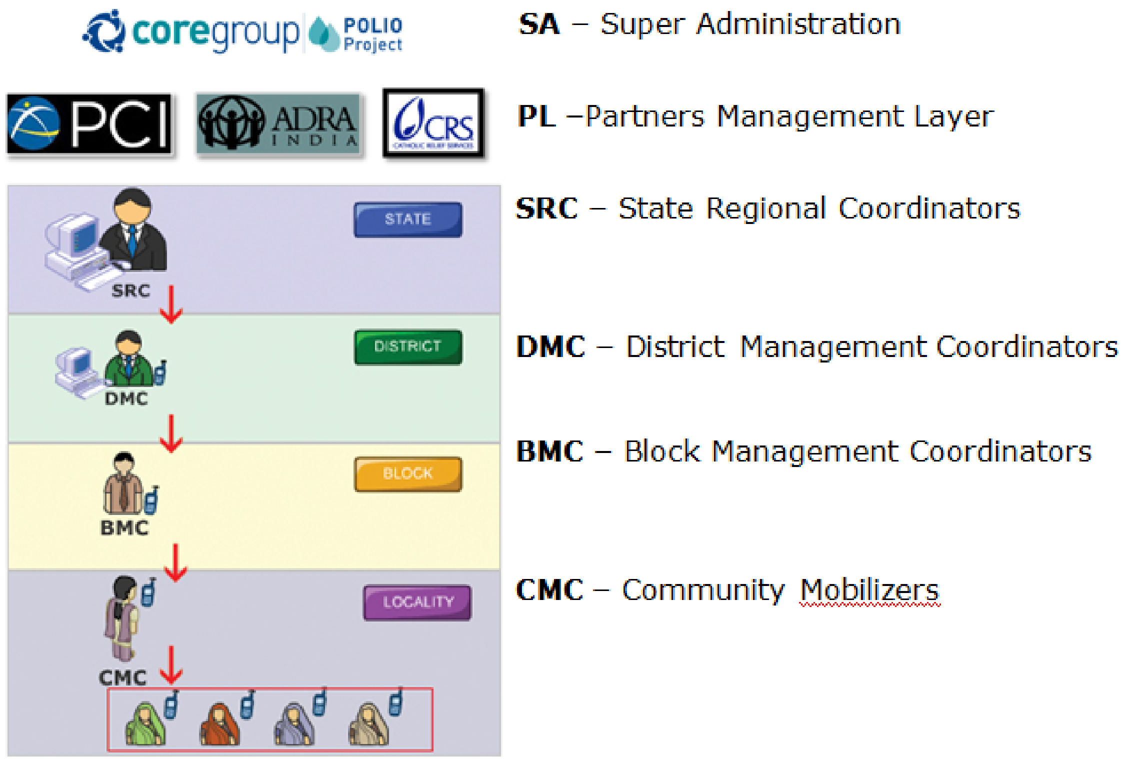


Fig.1.1 Implementation Structure

1. Community Mobilization Coordinators (CMCs)

There are 1325 salaried CMCs. Each CMC is responsible for mobilizing about 500 households, including holding community meetings, bringing local influential people to home visits, and holding health camps. Each CMC maintains immunization status records for all of the under five children in their assigned areas and for those participating in vaccination campaigns.

2. Block Mobilization Coordinators (BMCs)

There are 97 BMCs. Each BMC is responsible for social mobilization activities at the block level and overseeing and leading the CMCs.

3. District Mobilization Coordinators (DMCs)

There are 14 DMCs. Each DMC is responsible for supervising the BMCs and conducting social mobilization activities throughout the district.

CHAPTER 2

REQUIREMENT SPECIFICATIONS AND ANALYSIS

2.1 REQUIREMENT SPECIFICATION

It goes without saying that an accurate and through understanding of software requirements is essential to the success of .a software development effort. All further development like System Analysis, System Design, and Coding will depend on how accurate and well prepared the Requirement Specification is. Poorly analyzed and specified software will disappoint the user and bring grief to the developer, no matter how well designed and well coded the software is.

Requirement Specification appears to be a relatively simple task, but appearances are often deceiving. Chances of misinterpretation are very high, ambiguity is probable and communication gap between Customer and Developer is bound to bring confusions. Requirement specifications begin with a clear and concise heading stating in a sentence the task to be performed. Then requirement are described in a technical manner in precise statements. After the initial specification reports are received, they are analyzed and redefined through customer developer interaction. System Analysis follows to determine feasibility and profit loss analysis.

2.2 NEED FOR “M-HEALTH APPLICATIONS

India has a population of over a Billion people and has over 600 million mobile phones. Mobile technology creates practical and sustainable mobile solutions & services to enable social development by connecting them with people. Mobile solutions offer an effective means of bringing sustainable solution and services to people in the developing countries. With low-cost handsets and increasing penetration of mobile phone network globally, millions of peoples that that have never regular access to computer or fixed-line telephones now use mobile devices as daily tools for communication and data transfer. Today, there are 3.5 billion mobile phone users in world; more than 60% of them are in developing world. This explosion of mobile phone usage has the potential to service delivery on massive scale.

With time mobile device will serve as the prime platform of connected with the backend offices and systems of the world. The time is not far when Mobile applications will support all services such as m-Commerce, m-Health, m-Learning and over all sustainable development and commercial solutions.

2.3 ABOUT CONNECTED MOBILE GAMES AND LEARNING TOOLS

Today's changing the size of the devices and demands portability of devices. With the increase in the plethora of delivery devices and bandwidths, people are getting more and more mobile so there is no better medium of providing facility of application. Mobile devices are useful for learning, edutainment and access information at far distance through wireless. There is also a huge need of embedded games and application on cheap hand-held devices for to make devices faster.

Mobile learning is a new concept. ZMQ has identified it is a major learning tool of the already mobile devices like mobile telephones are in extensive use even in the remote villages. We believe that the learning solutions on mobile can enhance the living standard of not only the rich but also the poor. ZMQ is working on solutions on Childcare, women health, hygiene and micro-finance. Powerful solutions like m-tourism and virtual walks with m-Maps can be delivered.

2.4 IDENTIFICATION AND DESCRIPTION OF THE PROBLEM

The Existing system is not computerized and it is handled manually. It is very time consuming & need no. of people to do the jobs of the system.

- The existing System has many problems like, users have to do lots of paper work for data entry, transactions generation.
- The retrieval of data from the register is very tough task and every calculation on that data is not very simple as well as manual calculation is very slow in comparison of computerized system.
- Here in this case much time consumes, even a single mistake may create a problem.

Thus, I conclude that existing system not reliable for the organization. Hence it require a perfect computerized system.

The analyst has to identify the requirements by talking to these people and understanding their needs.

2.5 VARIOUS ASPECTS OF THE PROBLEM

2.5.1 Feasibility Study of the System

The feasibility study is carried out to test if the proposed system is worth being implemented. Given unlimited resources and infinite time, all projects are feasible. Unfortunately, such result and time are not possible in real life solutions. Hence, it becomes both necessary and prudent to evaluate the feasibility of the project at the earliest possible time in order to avoid unnecessary wastage of time, effort and professional embarrassment over an ill—conceived system. Feasibility study is a test of system proposed regarding its workability. Impact on the organization ability, to meet user who is familiar with the information system techniques.

Understand the part of the business or organization that will be involved or affected by the project, and are skilled in the system analysis and design process.

Three key considerations invoked in feasibility study are-economic, technical and behavioral feasibility.

2.5.1.1 Economic Feasibility

It is normal for every organization to choose a system development only if there is a gain with respect to time and cost overhead. In this case cost involved in development will be much less and time involved too is within constraints. So, software development is feasible.

2.5.1.2 Technical Feasibility

Technical Feasibility is the most difficult area to assess at the earlier stage of system development process. Therefore the process of analysis and definition of the proposed system was conducted in parallel with the assessment of technical feasibility. Since the misjudgment at this stage will be disastrous a cynical attitude was maintained throughout the evaluation of technical feasibility. This project is very small and will not require any additional technical environment in the company. Hence, it is technically feasible.

2.5.1.3 Behavioral Feasibility

The proposed system is behaviorally feasible, as it will provide a very user-friendly interface to play different games, select the option to be play and display the screen to play the game.

2.6 SOFTWARE PROCESS MODEL USED

To solve actual problems in an industry setting, software engineer or a team of engineers must incorporate a development strategy that encompasses the process, methods and tools layers and the generic phases of the software engineering. This strategy often referred to as a process model or a software-engineering paradigm. A process model is chosen based on the nature of the project and application, the methods and tools to be used, and the controls and deliverables that are required.

The nature of the given problem and the simplicity of the Linear Sequential Model forced the use of this Model in the development of this project. A brief discussion about this model is given below:

2.6.1 The Linear Sequential Model

The Linear Sequential Model is described through the diagram depicted below. Sometimes called the Classic Life Cycle or the Waterfall Model, the linear sequential model suggests a systematic, sequential approach to software development that begins at the system level and progress through analysis, design, coding testing, and maintenance. Modeled after the conventional engineering cycle, the linear sequential model encompasses following activities:

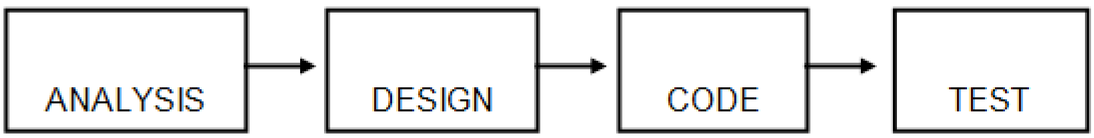


Fig: 2.1.The Linear Sequential Model

Analysis

The requirement gathering process is intensified and focused specifically on software. The information domain of the problem is identified end specified.

Design

Software design is actually a multi-step process that focuses on four attributes of a program; data structure, software architecture, interface representations and procedural (algorithmic details).

Code Generation

The design must be translated into a machine-readable form. The code generation step performs this task. If design is performed in detailed manner code generation can be performed mechanistically.

Testing

Once code has been generated, program testing begins. This testing process focuses on all the internal logic and the external behavior of the system.

2.6.2 System/Information Engineering and Modeling

This system view is essential when software must interface with other element such as hardware, people and databases. System engineering and analysis encompasses requirements gathering at the system level with a small amount of top-level analysis and design. Information engineering encompasses requirement gathering at the strategic business level and at the business area level.

2.7 REQUIREMENT ANALYSIS

As first steps in the analysis of the system, the end-user of the proposed system was met to get first hand information regarding their needs and wants. The user gave comprehensive information about the difficulties felt by him in doing his work. Ideas from both the sides were exchanged in order to get a standard and satisfactory system. Once overall goals were identified, a detailed analysis on the technology is done to make the system functional.

Software Requirement Specification (SRS) is the starting point of the software development activity. The objective of analysis of the problem is to answer the question: Exactly what must the system do? During system analysis the analyst attempts to develop a complete functional understanding of the proposed system. The document identifies a number of processes or functions that must be performed by the system.

There are mainly two parts of this phase:

1. Problem Analysis.
2. Requirement Specification and Review.

2.7.1 Requirement Analysis Elements

2.7.1 .1 Modeling

We created model to gain a better understanding of the actual entity to be built. Here entity is to be built is software, so it must be capable of modeling the information that the software transforms, the functions and sub-functions that enable the transformation to occurs, and the behavior of the system as the transformation is taking place.

2.7.1 .1.2 Functional Model

Software transforms information, and in order to accomplish this, it must perform at least three generic functions: input, processing, and output. The functional mode begins with a single context level model (i.e. the name of the software to be built). Over a series of iterations, more and more functional detail is provided, until a through delineation of all system functionality is represented.

2. 7.1 .1.2 Data Modeling

Data modeling defines primary data objects, composition of each data object, attributes of object, relationships between the objects and the processes.

2.7.1 .2 Data object, Attributes, and Relationship

The data model consists of three interrelated pieces of information: the data objects, the attributes that describe the data object, and the relationships that connect the data object to one another.

2.7.1 .2.1 Data Objects:

A data object is a representation of almost any composite information that must be understood by the software. By composite information, we mean something that has a number of different properties or attributes.

2.7.1 .2.2 Attributes:

Attributes define the properties of a data object.

2.7.1 .2.3 Relationships:

Data object are connected to one another in a variety of different ways. We can define a set of object object-relationship pairs that define the relevant relationships. Object relationship pairs are bi-directional.

2.8 HARDWARE AND SOFTWARE REQUIREMENT

2.8.1 Hardware Requirements:

Pentium CPU @ 450 MHz
256 MB RAM
512 KB Catch
80 GB SCSI Hard Disk
SVGA 1024 x 768 resolution

2.8.2 Software Requirements:

OS—Windows XP or above.
Platform: NetBeans
Database: SQL Server 2005
GUI - J2ME Wireless Toolkit.

2.9 THE OBJECT MODEL

Object-oriented technology is build upon a sound engineering foundation whose elements are collectively called the *object model.* The object model encompasses the principals of abstraction, encapsulation, modularity, hierarchy, concurrency and persistence.

Object Oriented programming: Object Oriented Programming is a method of implementation in which programs are organized as cooperative collections of objects, each of which represents an instance of some class, and whose classes are all members of a hierarchy of classes united via inheritance relationships.

Object-Oriented Design: Object-Oriented Design is a method of design encompassing the process of object-oriented decomposition and a notation for depicting logical and physical as well as static and dynamic models of the system under design.

Object-Oriented Analysis: Object Oriented Analysis is a method of analysis that examines requirements from the perspective of the classes and objects found in the vocabulary of the problem domain.

CHAPTER 3

SYSTEM ANALYSIS

3.1 DESIGN

3.1.1 System Design

Once the software requirements have been analyzed and specified, software design is the first of the three technical activities- designs, code generation, and testing- that are required to build and verify the software. The design step produces a data design, an architectural design, an interface design, and a procedural design. The data design transforms the information domain model created during analysis into data structures that will be required to implement the software. The objects, relationships and detailed data content provided the basis for the data design activity.

The **architectural design** defines the relationship among major structural elements of the program.

The **interface design** describes how the software communicates within itself, to systems that inter-operate with it and with humans who use it. An interface implies a flow of information.

The **procedurals design** transforms structural elements of the program architecture into; a procedural description of software components.

During design, we make decisions that will ultimately affect the success of software construction. The importance of design can be stated with a single word “quality”.

3.1.1.1 Study design

This study is a secondary analysis of data originally collected for the purpose of program management. The original data include information about CMC areas and non-CMC areas in all blocks that the CGPP works in (e.g., there was no sampling of sites within the CGPP program area). Given the type of data available for this paper, we chose a quasi-experimental uninterrupted time series analysis design with a non-equivalent comparison group. This involves comparing CMC and non-CMC areas separately over time (each of the national immunization days.

Program/intervention areas (CMC) and comparison areas (non-CMC) were not selected at random. CMC areas are a group of villages within a block that have been purposively selected because they have the most difficulty achieving vaccination program targets. CMC areas have a higher proportion of families that are resistant to allowing their children to be vaccinated with polio vaccine than non-CMC areas. Therefore, we would expect---in the absence of additional CGPP social mobilization efforts---that vaccination performance would be lower, on average, in CMC areas compared to non-CMC areas. However, if performance in CMC areas approaches or surpasses the performance in non-CMC areas, we might judge the social mobilization efforts as having had positive effects.

3.1.1.2 Description of data

The original data represent programmatic monitoring data collected by CMCs during the course of their ongoing work as follows. In both CMC and non-CMC areas, government vaccinators from the Ministry of Health, supervised by their block level managers, tally standard information about the SIA (such as the number of children under five years who were vaccinated or the number of households with children under five remaining to be vaccinated). In CMC areas, the CMC is simultaneously collecting the same data. The CMCs maintain a detailed register that records the following: the number of children vaccinated at booths; the number of households visited after the booth day; and, the eventual immunization outcome at each house. Joint government and SM Net teams (BMCs and their block level government counterparts from the Ministry of Health) sit together and review vaccinators' tally sheets each night of the campaign. In addition, independent monitoring teams survey a percentage of houses after each SIA.

Each month, the CGPP CMC compiles her data and delivers a hard copy report to her supervisor, the BMC. The BMC compiles CMC data and submits it to the DMC who then sends a computerized report to the SRCs and the CGPP Secretariat. Data for non-CMC areas of each block were obtained by subtracting numbers for the CMC areas from the block's total (provided by the government). The CGPP Secretariat maintains a central database of these data. These data were then compiled and cleaned for this secondary data analysis.

The CGPP tracks three important indicators of social mobilization performance during SIAs: (1) Booth Coverage; (2) Percent of "X" households converted to "P" during both A and B team activities; and, (3) Percent total "X" households converted to "P" during B team activities (among those "X" households remaining after A team activities). These three indicators are considered sensitive to the social mobilization efforts of the CGPP CMCs and important milestones for vaccinating every child that needs it. Booth coverage is the proportion of eligible children that were vaccinated at vaccination booths. The denominator is the total number of children vaccinated during the previous SIA (at booths and during house-to-house visits by vaccinators), and therefore varies between SIAs. CMC activity should increase Booth Coverage by motivating families to take their children to vaccination booths located in their own communities. The definition of an "X" household is discussed above. CMC activity should also increase the Percent of "X" households converted to "P." The reasons for a household to have unvaccinated children (to be an "X" household) could include overt refusals by caregivers, sick children, children not present during vaccination, or locked houses. The CMC works hard to convince resistant parents and parents of sick children to get their children vaccinated.

The CGPP uses a database to maintain information about vaccination activities during SIAs. Each record in the database contains information about vaccination performance during a single SIA (organized by month) in one block. There is a separate record/row each representing the CMC areas in a block and the non-CMC areas in the same block. CMC areas cover roughly one third of the entire block while non-CMCs areas account for the remaining two-thirds. Record/rows in the database include the following information: district in which the SIA was conducted; block in which the SIA was conducted; whether the record represents a CMC or non-CMC area; the month and year of the SIA; the estimated number of children under age of five years living in the areas that the record represents; the three performance indicators described above; and, other data needed to calculate the three indicators.

3.1.2 Data Design

The data design is first of four design activities that are conducted during the software engineering. The impact of data structure on program structure and procedural complexity causes data design to have a profound influence on software quality.

3. 1.3Architectural Design

The primary objective of architectural-design is to develop a modular program structure and represent the control relationships between modules. In addition architectural design melds program structure and the data structure, defining an interface that enables the data to flow throughout the program.

3. 1.4 Interface Design

Interface design focuses on the three areas of concern:

1. The design of interface between software modules.
2. The design of interfaces between the software and other non-human producers and consumers of information.
3. The design of interface between a human and the computer.

The help is provided as a manual to the user within the system. Online help is not necessary for this system as the users are mostly program monitors and the interface is very simple.

The interface is very helpful, friendly and easy, with respect to the error warning and the online massage. On each action the system responds, with warning and the error messages, where necessary.

3.1.5 Procedural Design

Procedural design occurs after data, architectural, and interface designs; have been established. The procedural design specifies the algorithmic detail of each of the function (module) in PDL (Process Design Language).

3.2 SOFTWARE DESIGN DOCUMENT

3.2.1 Introduction

The Community Mobilization Coordinator (CMC) interacts with families and community members at the village level. As the backbone of the SM net, s/he is assigned responsibility for mobilizing about 500 households in both a rural or urban area, and keeps records of the immunization status of all 0-5 year children in those households. During SIA (Supplementary Immunization Activity) rounds, CMCs assist vaccinators in setting up vaccination booths, organize groups of child mobilizers (*Bulawwa tollies*) and arrange for mosque and/or temple announcements. CMCs also do the following: accompany vaccinator teams to all the houses; work to convince families with an unvaccinated child (called an 'X' household) to allow their child to be vaccinated (converting an 'X' household to 'P' (denoting a house where all eligible children are vaccinated against polio); and, accompany persons of influence (influencers) during follow-up activities.

In between the SIA Rounds, the CMC carries out activities aimed at increasing OPV coverage. S/he visits houses, talking to mothers and other caregivers, dispelling their doubts or rumors about the vaccine. S/he holds mothers meetings to discuss their children's health and explains prevention and management of common illnesses. One of the most interesting activities that s/he conducts is harnessing the potential of school children. S/he conducts 'Polio classes' at schools in her/his area. In these classes, she uses various methods to get the children interested in becoming a part of the polio campaign---from poetry and painting competitions on the Polio theme, to rallies. A few children are then selected to come together as 'Bullawa tollies' (Literal translation = Calling gangs) who mobilize mothers to bring their children to the booths during SIAs.

In India, the Block is the smallest administrative unit and is made up of 100-150 villages. Within the SM Net, the Block Mobilization Coordinator (BMC) oversees social mobilization activities during (and in between) SIA rounds through supervision and mentoring of the CMCs working in the block. The BMC reports to the District Mobilization Coordinator. H/she is responsible for the following activities: capacity building of all CMCs through training; data collection and collation by CMCs; building partnerships with the Medical officer in charge of the Block Primary Health Center and other stakeholders; ensuring that routine immunizations are conducted in hard to reach areas; conducting inter-personal communications (IPC) sessions; participating in their block task force and other related meetings; organizing health camps; and, monitoring CMC vaccination booths and house-to-house vaccination visits.

The District Mobilization Coordinator (DMC) is in charge of social mobilization activities in all the CORE blocks of the district. S/He is responsible for the following activities: compilation of all block level data that are sent to the Secretariat; training; participation in District Task Force meetings; developing the joint SM Net District Communication Plan; building strong partnerships with the district government officials and other stakeholders; and, ensuring routine immunization sessions are conducted in hard to reach areas. During SIAs, the DMC monitors the quality of vaccination activities at CMC vaccination booths and during house-to-house visits. In addition, h/she updates records about the SIA and provides feedback to the Chief Medical officer, and medical officers in charge of the health centers. The District Underserved Coordinator (DUC) is in charge of planning and implementation of activities in underserved areas in the district, such as liaison with religious leaders and religious institutes and ensuring mosque announcements. The Sub Regional Coordinator (SRC) is an employee of one of the CGPP partner organizations and oversees activities in multiple districts. S/He collaborates with local government departments, non-governmental organizations, religious institutions, and other polio partners.

3.2.2 How Vaccination Is Done During SIAs In SM Net CMC Areas

Vaccination during SIAs in SM Net areas is conducted in the following way. On Day 1 of the SIA, almost always a Sunday, fixed booths are set up where the vaccination teams are stationed. The team consists of three persons, and the third person is hired from the same community to mobilize children from houses. In CMC areas, CMCs accompany team members and mobilize children by making *bulawwa tollies* of older children from the same neighborhood (as described above). From Day 2 through Day 5 or 6, vaccination teams, called A Teams, visit about a hundred households each day to document vaccination of all eligible children. Once the vaccination team has visited all assigned households on a particular day, they count the number of households that have not had all eligible children vaccinated. These are called "X" households. The vaccination team then revisits all the "X" households to vaccinate children. The CMC accompanies the vaccination team during visits to "X" households and uses various methods to convince families to vaccinate all eligible children. If an "X" household allows all eligible children to be vaccinated, then we say that an "X" household has been converted to "P." For example, if a family refuses to allow their child to receive vaccine, the CMC may arrange for an influential person in the community to come and speak with the family and encourage vaccination. This ends the A Team effort on a particular day for the team's assigned households. There might remain a number of "X" households, however, in the section of households that the A Team passed through that day.

On the day after the A Team has passed through a section of houses, another team, called B teams, visits the households still labeled "X" after the A Team's efforts. This is the last chance during the SIA to vaccinate eligible children in "X" households through B team activities. The CMC accompanies B team members and visits all "X" houses remaining after A team to communicate to the families, the importance of OPV, and also, if required, to arrange influencer's visit. At the end of the B team activities, there might still remain a number of "X" households.

The Indian Academy of Pediatrics (IAP) has praised the efforts of SMNet in western UP for reducing the number of "X" houses during house to house immunization, increased booth coverage, and led to a reduction in 'resistant' households. The IAP additionally encouraged the SM Net to continue its social mobilization efforts in UP. This is important because social resistance to polio vaccine remains a key barrier to eradication in western UP. As the social mobilization efforts of the SM Net in western UP continue, it would be useful to more rigorously evaluate the outcomes of these efforts.

The objective of this project is to digitalize the polio activity and to vaccinate children through SM Net as SM Net areas are better than in non-SM Net areas within the CGPP program.

3.2.3 Purpose

ZMQ states that, "With low-cost handsets and increasing penetration of mobile phone networks globally, millions of people that never had regular access to computers or fixed-line telephones now use mobile devices as daily tools for communication and data transfer. Today, there are 3.5 billion mobile phone users in the world; more than 65% of them are in the developing world." According to ZMQ, in spite of the huge penetration of mobile networks, development programmes frequently suffer from inadequate access in terms of content, applications, solutions, and services. "M4D invites social development organizations to partner in this initiative to implement new programs or scale their existing programs using mobile technology."

ZMQ is a winner of the United Nations Development Programme (UNDP)'s World Business and Development Award 2008 in recognition of its works in improving lives of the world's most disadvantaged people and efforts aimed at reaching the UN Millennium Development Goals through technology.

3.2.4 Scope

- Awareness and communication: mobile phone games, embedded applications, 1-way short messaging service (SMS), 2-way, interactive SMS response system, and interactive voice response (IVR) systems.
- M-training for field workers: m-Learning platform for field officers (health workers, research scholars, etc.) with back-end monitoring and evaluation system.
- Remote data collection: SMS with a back-end database integration and mobile client-server application for audio, video, and image capture.
- Remote monitoring: SMS alerts, IVR solutions, and client-server monitoring applications and surveillance tools.
- Mobile enterprise solutions: mobile client-server application for care, support, and treatment.
- Payments and transactions: secure m-payment system, mobile client-server transaction solutions, m-banking and SMS-based voucher validation and scratch-cards.

The scope of this document is throughout the Life Cycle of the product. Any change in the design leads to change in the subsequent phase of the Life Cycle.

3.2.5 Overall Description: Design perspective

The analysis model provides a detailed understanding of requirements. Most important it imposes a structure of the system that we should strive to preserve as much as possible when the system is designed.

Within the design model the design classes and their objects realize use cases. Object oriented design involves two main design steps:

- Use Case Design
- Class Design

3.2.5.1 Use Case Design

We begin our design by designing by each of the use case. The input for the use case design is the requirement model (use case model and) Analysis Model.

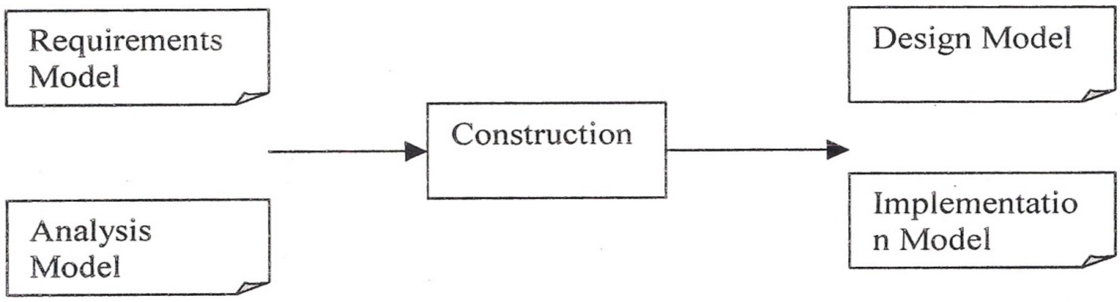


Fig. 3.1: The Input and Output models of construction.

In use case design we do the following:

1: We model the first level design collaboration diagram from analysis collaboration diagram using one to one mapping between analysis classes to the design classes.

2: Now we find the candidate design classes.

- Choose the candidate class from the first level design collaboration diagram.
- Discuss the implementation platform and read those requirements of the use case that need to be discussed at that time. Identify the more candidate classes.
- Identify the subsystem (Middleware) that is required to design the use case as per the Implementation platform.
- Identify the active classes. The active classes are those classes that have their own control of thread.

3: when the entire candidate classes have been identified and their relationships have been resolved, we prepare the second level design collaboration diagram.

4: this second level design collaboration diagram will model all the design classes and their collaboration to carry out the use case goal.

5: we now prepare the sequence diagram. Sequence diagram is an interaction diagram that shows how the use case goal is carried out by the interaction through the message between the design objects the message is the service requested by the invoking object. The sequence of action is shown on the time line basis. An actor and the purpose of actor being served by the interaction among the design objects initiate the use case.

6: After this step any inheritance hierarchy, if exist can be described among the classes. The second step in the design is to design each of the design classes identified above.

3.2.5.2 Class Design

Design of a class involves the following:

1: design of attributes: attributes are designed as follows:

- Identify the information that is required to be modeled in this class. The information is identified by the analysis class description.
- If this class interacts with any other class to carry out its task, then make this second class as the attribute of the first class.

2: Design of methods: Methods are designed as follows:

- Identify the message in sequence diagram passed by one object to another object. This message to another object is the service required by the first object. This message is designed in one more methods. Typically the message itself is the method name of the object.
- Model any other method that is required to perform the responsibility as private methods used by method.
- In this way find out all the messages toward these objects, which other objects are dependent. These all messages are the methods of the class. The parameter to the method is the input provided by object requesting service and return values are what the requesting object is desired.

3.2.5.3 Design for class application:

Hence the design model of the Search Mail System can be described as:

- Design Model for Embedded Games for Mobile Devices Term used within the document.

- End User

3.2.6 Design of Use Case

1. First Level Collaboration Diagram of INTRFACE

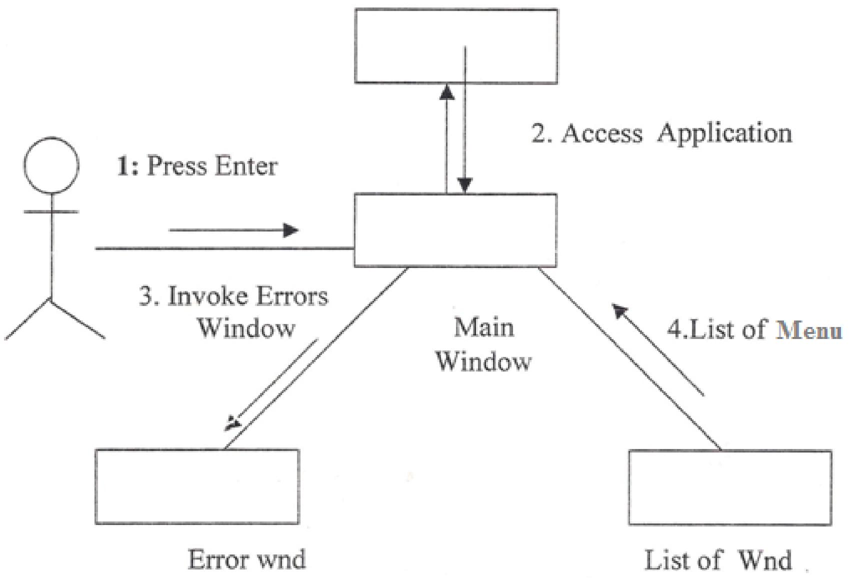


Fig. 3.2 Collaboration Diagram of INTERFACE

2. Sequence Diagram

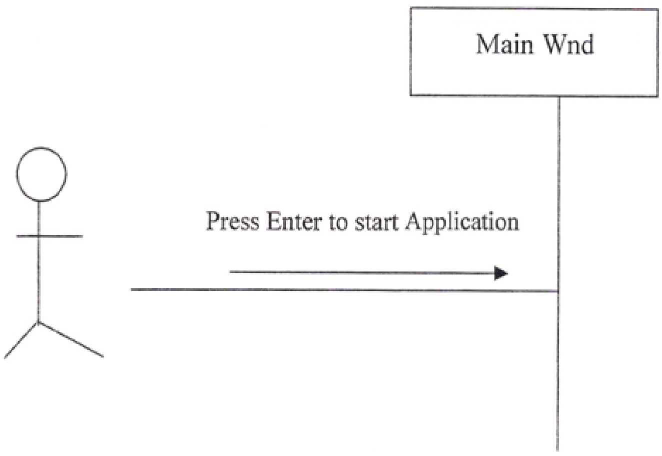


Fig. 3.3 Sequence Diagram of INTERFACE

II. Action Performed

- A Main Window will display containing the list of different menu of select any option from the list.

1 Design use case INTERFACE

I. First level Collaboration Diagram INTERFACE

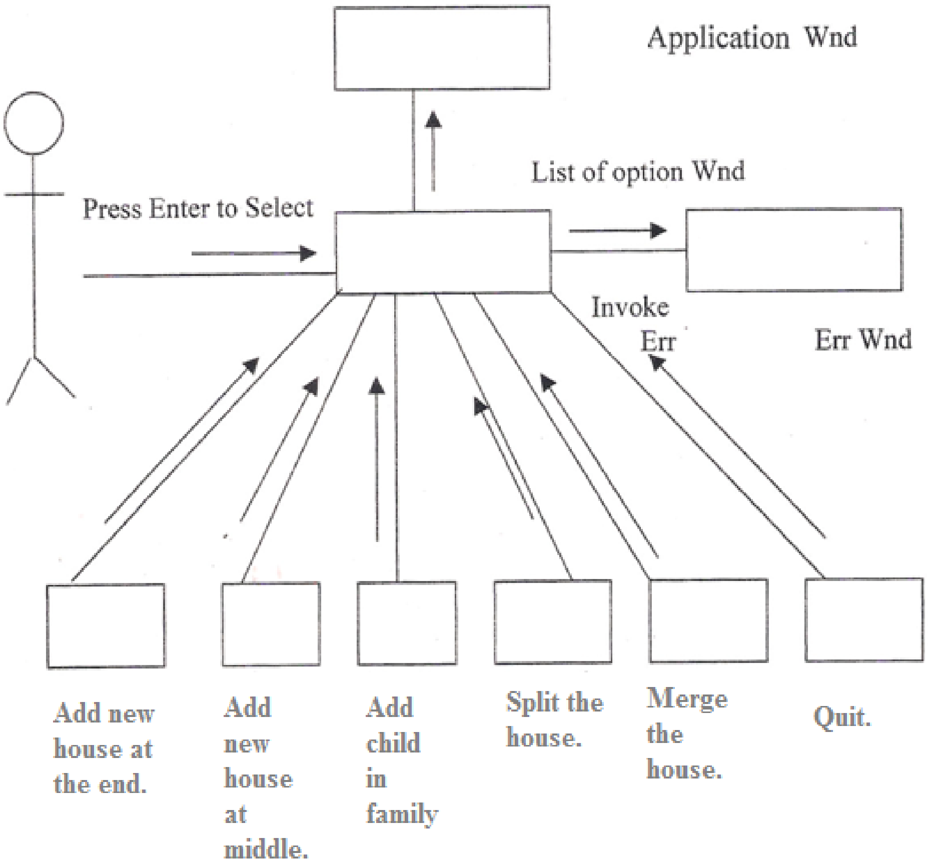


Fig. 3.4 First level Collaboration Diagram of INTERFACE

II Sequence Diagram:

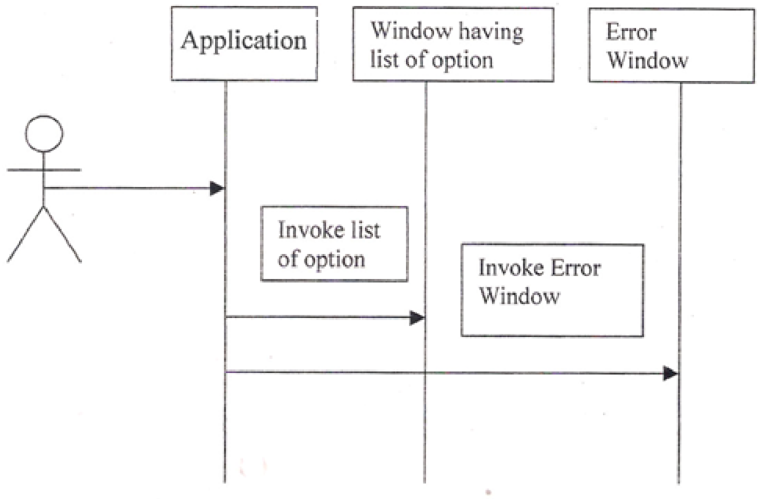


Fig. 3.5 Sequence Diagram of INTERFACE.

III Action Performed

- A list of option will display containing the list of option about the Application.

- The user pressed enter to begin the application
- A page containing the links for any option will display.

I. First level Collaboration Diagram of INTERFACE

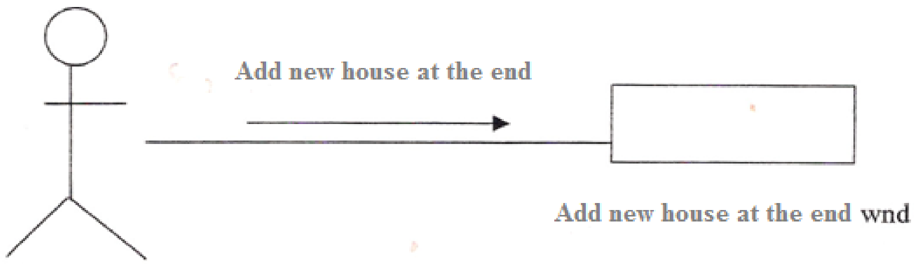


Fig. 3.6 Collaboration Diagram of house at the end

II Sequence Diagram

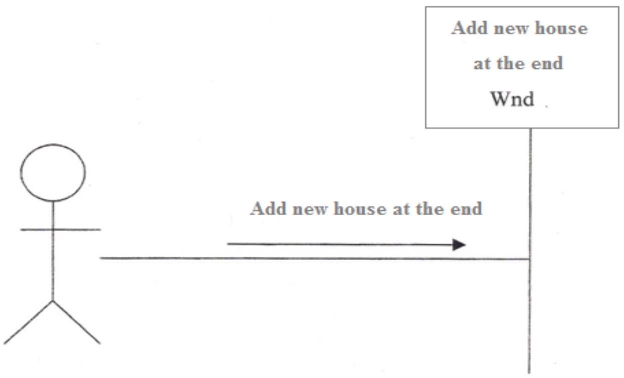


Fig. 3.7 Sequence Diagram of house at the end

I. First level Collaboration Diagram of INTERFACE

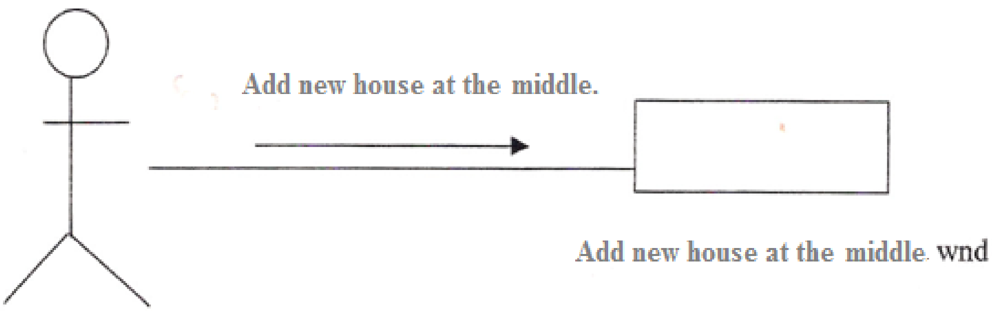


Fig 3.8 Collaboration Diagram of house at the middle

II Sequence Diagram

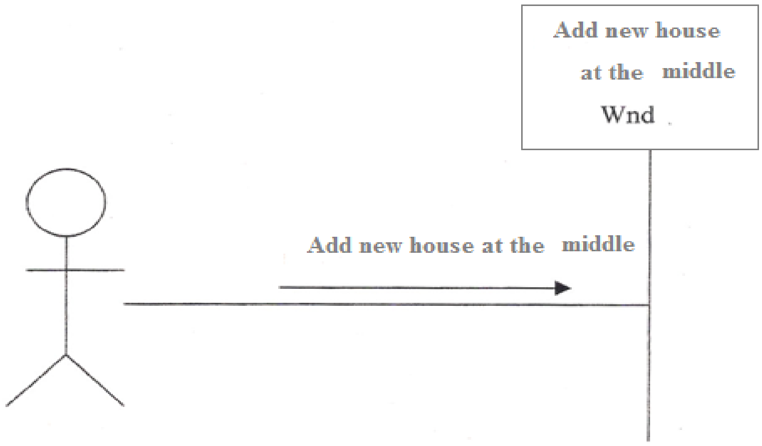


Fig. 3.9 Sequence Diagram of house at the middle

I. First level Collaboration Diagram of INTERFACE

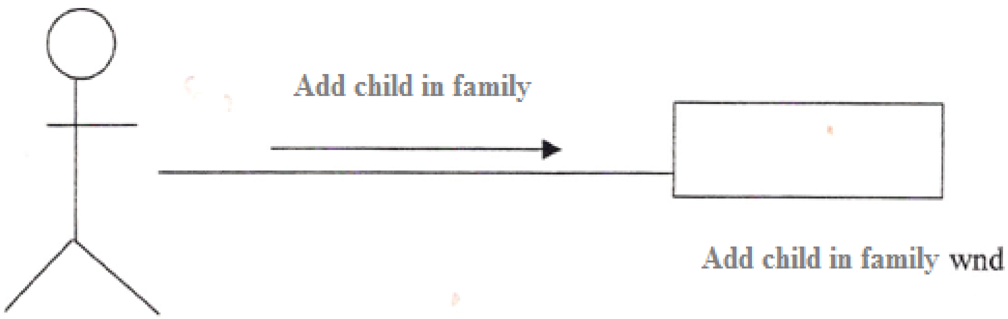


Fig. 3.10 Collaboration Diagram of child in the family

II Sequence Diagram

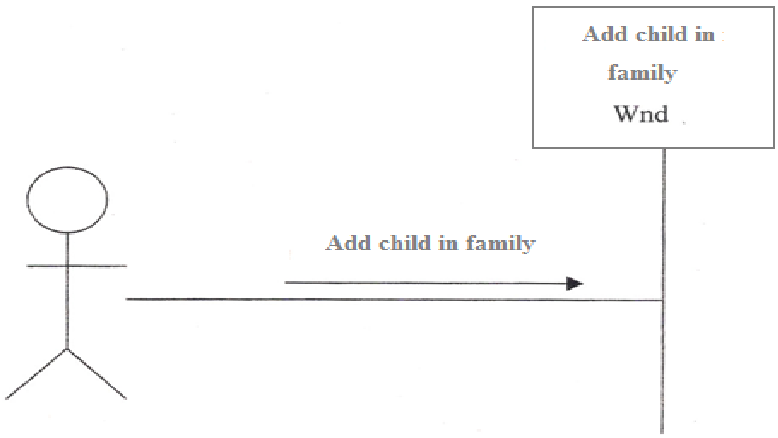


Fig. 3.11 Sequence Diagram of child in the family

I. First level Collaboration Diagram of INTERFACE

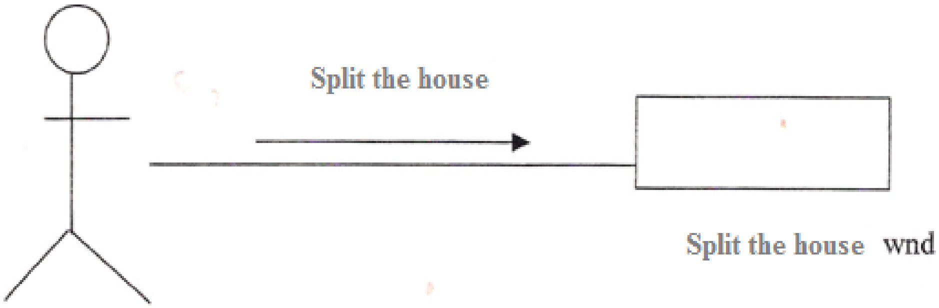


Fig. 3.12 Collaboration Diagram of split the house

II Sequence Diagram

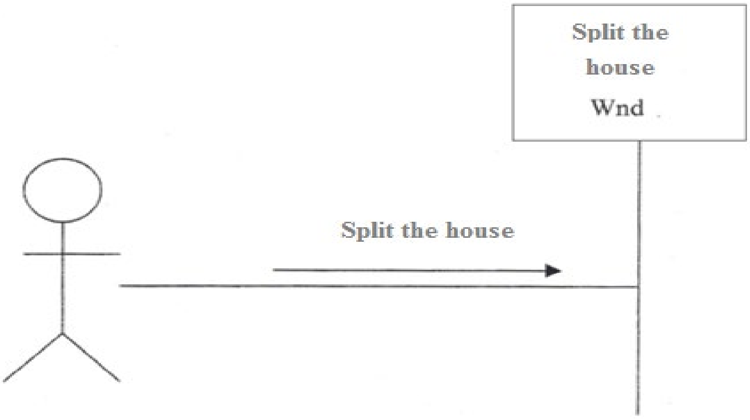


Fig. 3.13: Sequence Diagram of split the house

I. First level Collaboration Diagram of INTERFACE

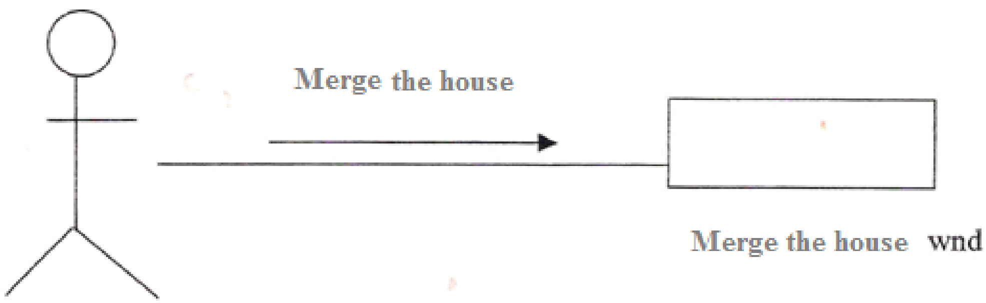


Fig. 3.14 Collaboration Diagram of Merge the house

II Sequence Diagram

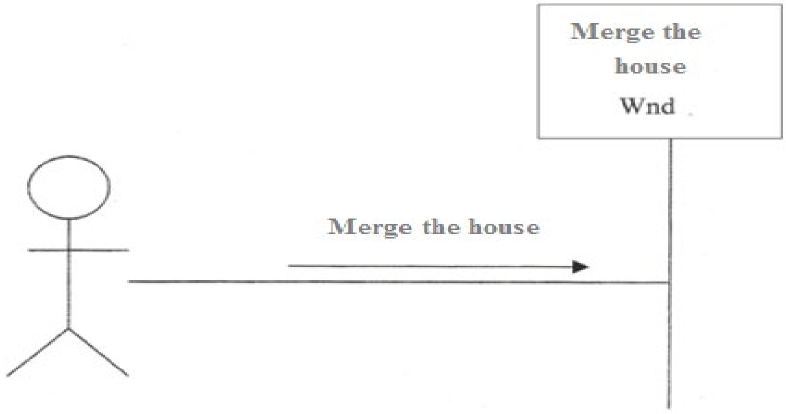


Fig. 3.15 Sequence Diagram of Merge the house

I. First level Collaboration Diagram of INTERFACE

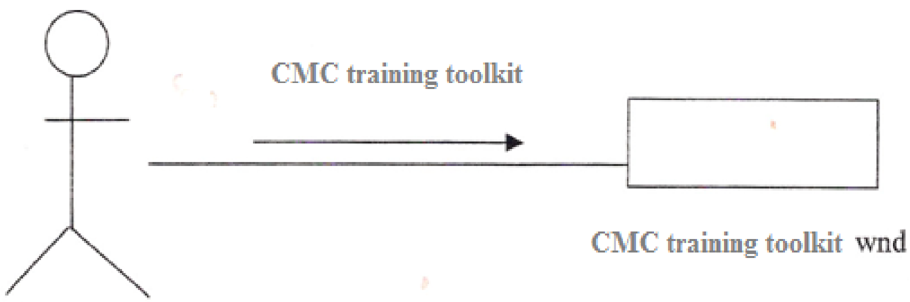


Fig. 3.16 Collaboration Diagram of CMC toolkit

II Sequence Diagram

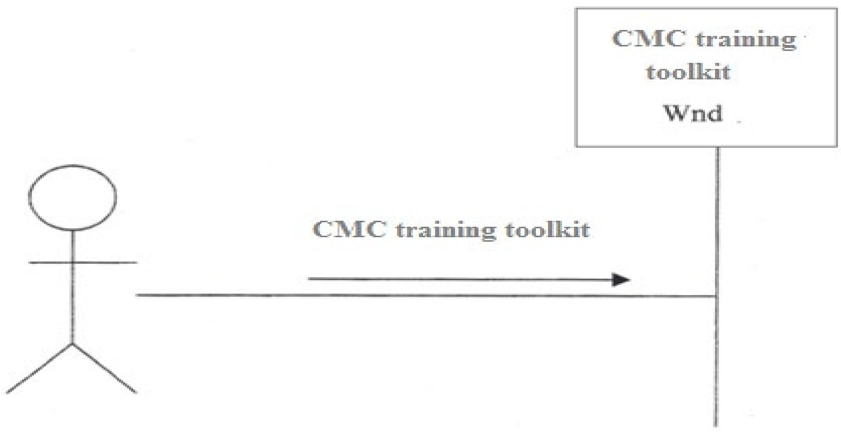


Fig. 3.17: Sequence Diagram of CMC toolkit

I. Flow Diagram of CMC training toolkit INTERFACE

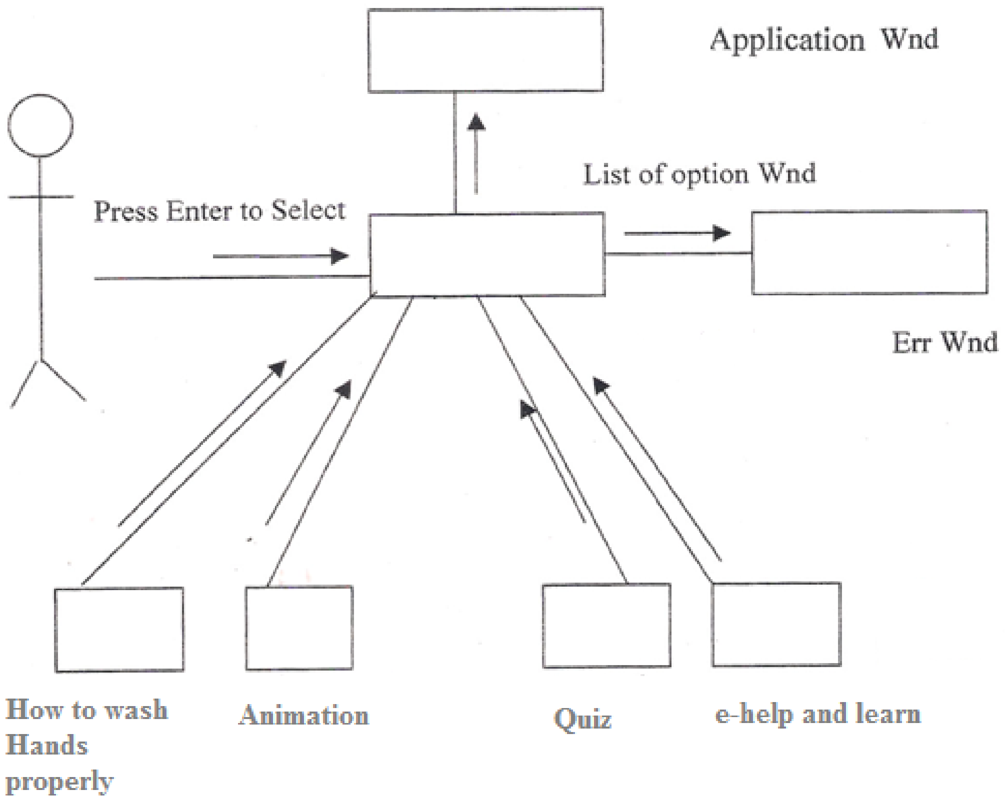


Fig. 3.18 Flow Diagram of CMC training toolkit INTERFACE

3.2.7 Design Model

3.2.7.1 Introduction

The Design Model describes the physical realization of use cases by focusing on functional and non-functional requirements, together with other constraints related to the implementation environment and impact of the system's implementation. Within the Design Model use design windows forms and their objects realize cases. We begin our design by designing the use cases. The use case design has the following purpose.

- Identify the design windows form and or/ subsystems whose instances are needed to perform the use case's flow or events.
- Distribute the behavior of the use case to the interacting design objects and/ or to the participating subsystems.

Define the design windows forms and/or subsystems and their interfaces according to the Software Requirements Specification documents and their operations.

3.2.7.2 Identify the Participating Canvas Screen

From the use case "realization-analysis" involving interaction among interface objects and control objects we have come to the following design Canvas Screen.

3.2.7.3 Description of the Canvas Screen

1. Main Menu
    (a) Add new house at the end
    (b) Add new house in the middle
    (c) Add child in family
    (d) Split the family
    (e) Merge the house
2. CMC Training Kit

3.2.7.4 Flow of Events

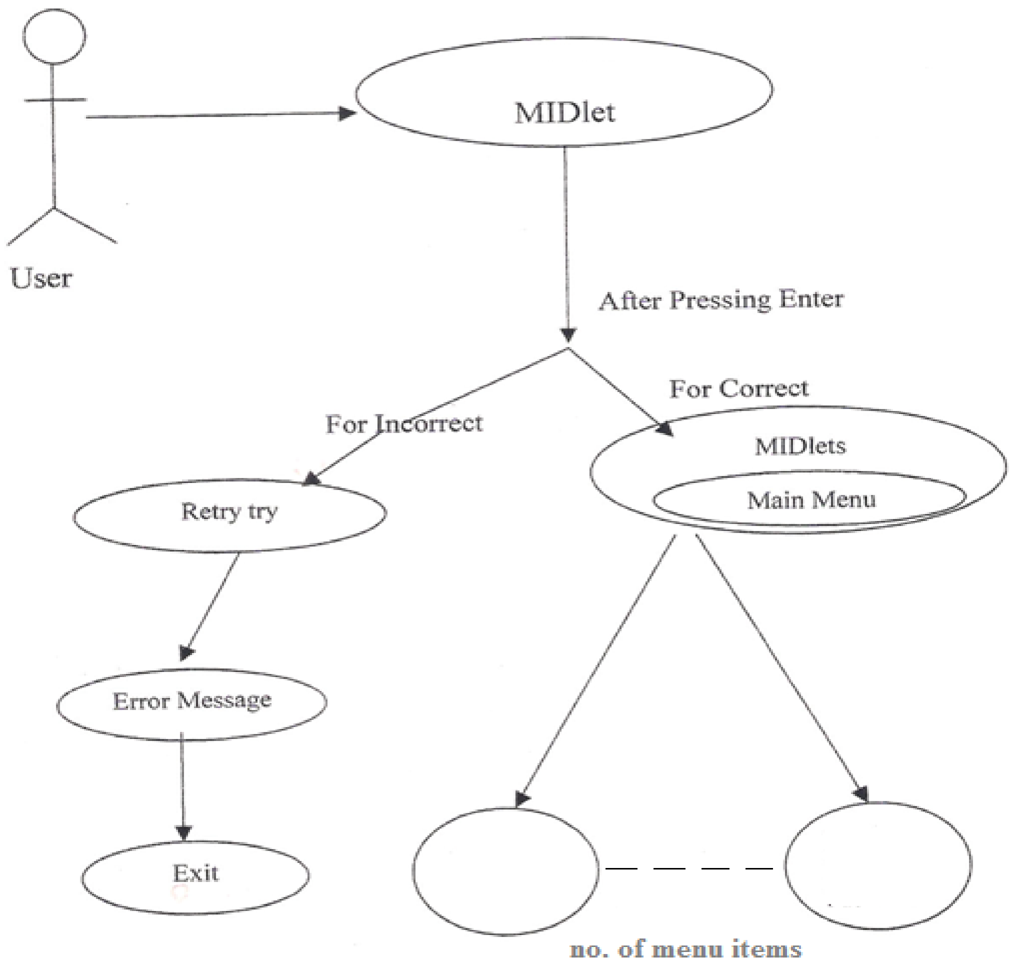


Fig. 3.19: Flow of Event

CHAPTER 4

DETAILED DESIGN

4.1 INTRODUCTION TO DETAILED DESIGN

During detailed design in the internal logic of each of the modules specified in their system is decided. During this phase further details of data structures and algorithms design of each module is specified. The logic of .the module is usually specified in high-level design description language, which is independent of the target language, in which the software is eventually implemented. All the database tables are specified in this phase.

4.1.1 Splash

The Screen, which is visible at the start of an application, is termed as ***Splash*** Screen. This Screen shows a brief idea of what the Application can be like.

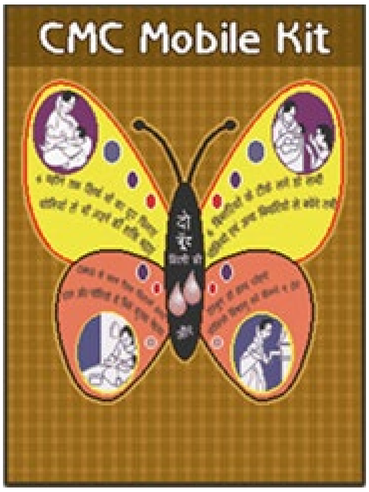


Fig. 4.1 Screen visible at the start of an application

Note: Screen will be visible from starting of the application. Once the user exits the Splash, it will not be visible unless he opens the same application again by closing it.

4.1.2 Login

The Second screen is the Login Screen. In This screen user enter their login id as a login Id and password that is provided by the company. After fill their number and password user select 'login' command to enter their menu screen. To exit from the Login Screen, the user has to select 'clear' command.

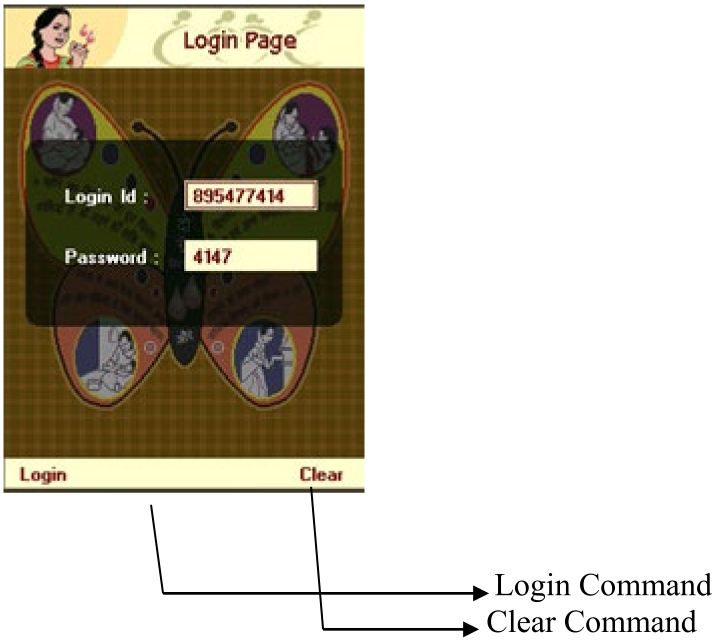


Fig. 4.2 Login Screen of application

4.1.3 Menus

The Third screen is the CMC Menu. This screen provides the options available for the cmc to interact.

Fig. 4.3 After Login Screen menu available for CMC

CMC can scroll through the menu by selecting 'Up (2 key)' and 'Down (8 key)'. To select the menu option, the user has to select 'Fire (5 key)'. To exit from the Application, the user has to select 'Exit' command. To exit from the Menu Screen, the user has to select 'Back' command.

4.1.4 Remote CMC Channel

When a health worker clicks CMC channel then he/she gets following menus:

1. How to wash hands properly

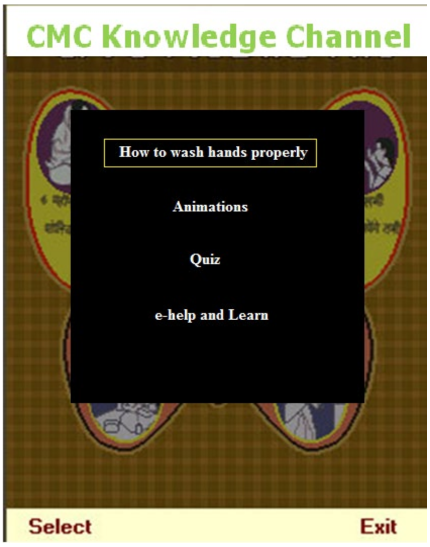


Fig. 4.4 Training CMC How to wash hands properly

Steps to show how to wash hands properly:

Fig. 4.5 Steps to show how to wash hands properly

Fig. 4.6 Steps to show how to wash hands properly

2. **Animations**

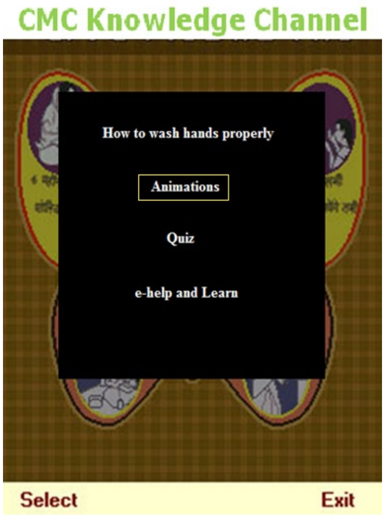


Fig. 4.7 Training CMC by Animation how to wash hands properly

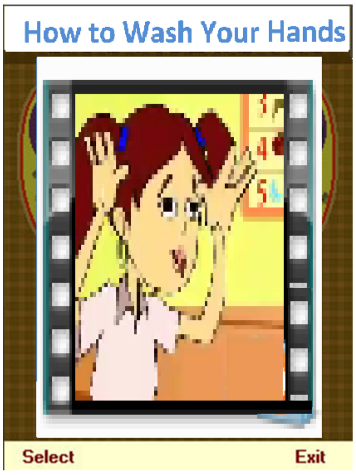


Fig. 4.8 Animation

3. **Quiz**

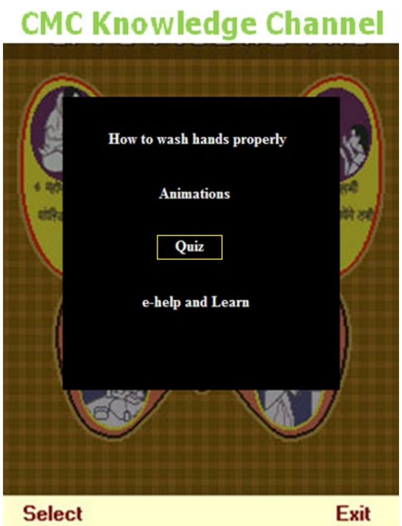


Fig. 4.9 CMC Quiz option

CMC taking quiz

Fig. 4.10 Steps to show CMC taking quiz

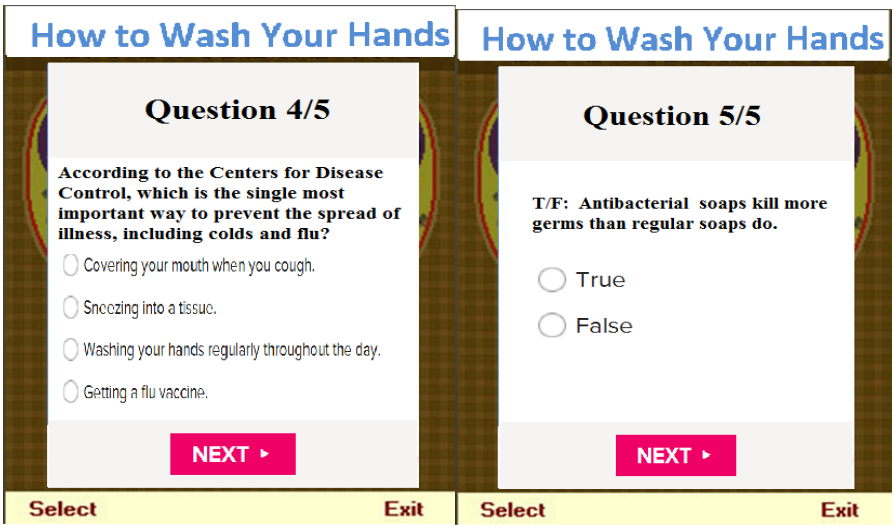


Fig. 4.11 Steps to show CMC taking quiz

4.1.5 General Information

The General Information screen is activated, when the user selects the 'General Information' option in menu. This screen displays personal information of the user: Name, Phone number, address of CMC. The user can go back to menu screen by selecting Back command and exit the application by selecting Exit command.

4.1.6 New Registration

As soon as a CMC visit in community and find a house with a child of less than five years of age or any pregnant women then it makes a new registration in her tool kit.

Fig. 4.12 Add new house at end

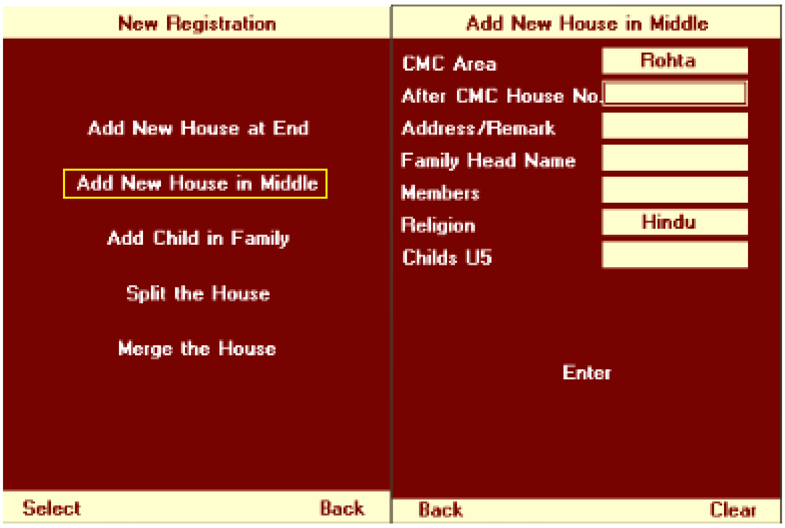


Fig. 4.13 Add new house in middle

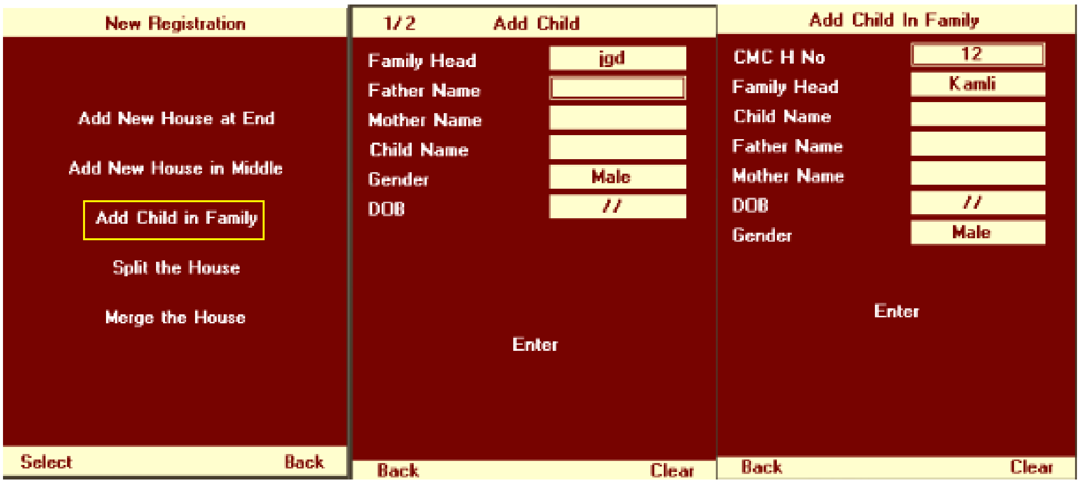


Fig. 4.14 adding child in a family

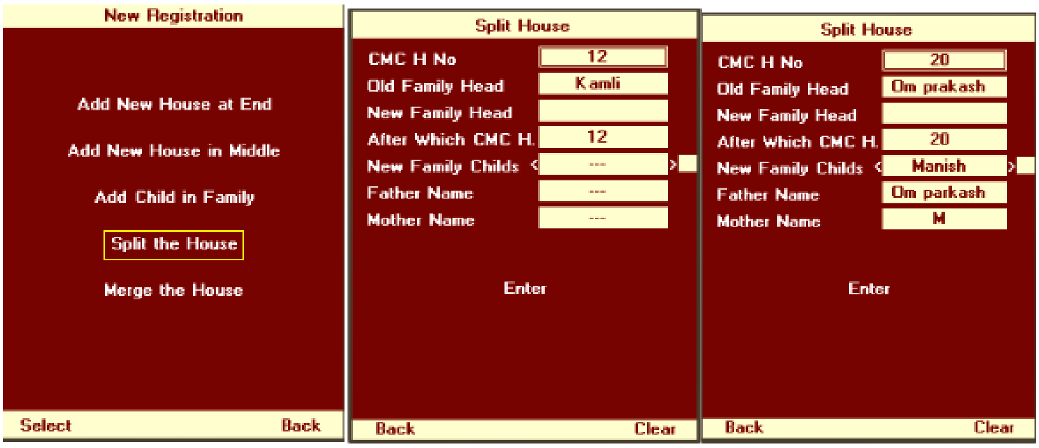
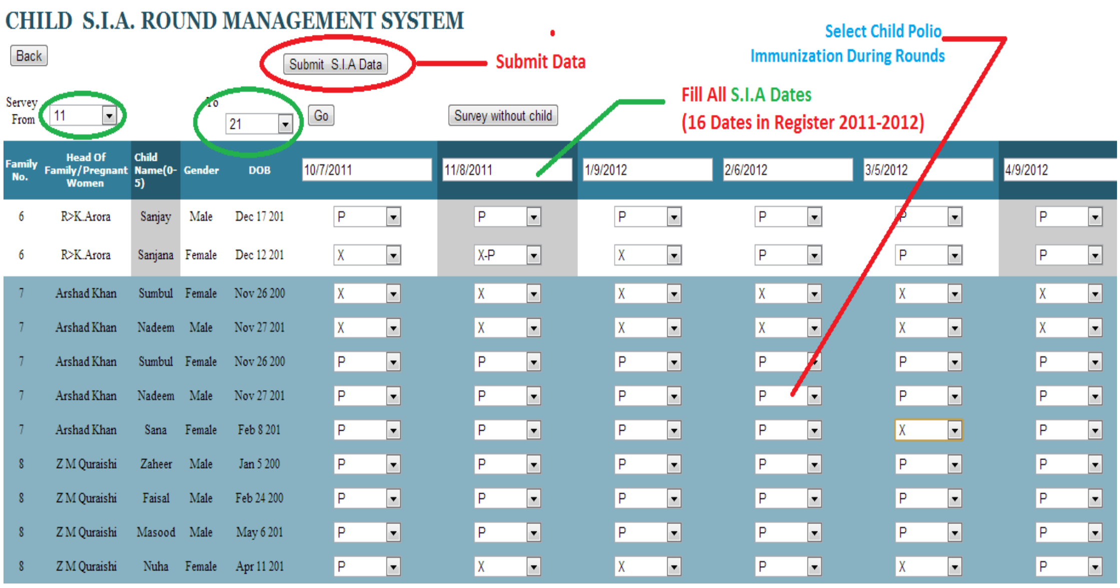


Fig. 4.15 splitting the house

Fig. 4.16 merging the house

4.1.7 Reporting system

At last reporting is done. It is done at every level and all health workers including BMC and DMC manage such entries which represents database entries.



Family No.	Head Of Family/Pregnant Women	Child Name(0-5)	Gender	DOB	10/7/2011	11/8/2011	1/9/2012	2/6/2012	3/5/2012	4/9/2012
6	R>K.Arora	Sanjay	Male	Dec 17 201	P	P	P	P	P	P
6	R>K.Arora	Sanjana	Female	Dec 12 201	X	X-P	X	P	P	P
7	Arshad Khan	Sumbul	Female	Nov 26 200	X	X	X	X	X	X
7	Arshad Khan	Nadeem	Male	Nov 27 201	X	X	X	X	X	X
7	Arshad Khan	Sumbul	Female	Nov 26 200	P	P	P	P	P	P
7	Arshad Khan	Nadeem	Male	Nov 27 201	P	P	P	P	P	P
7	Arshad Khan	Sana	Female	Feb 8 201	P	P	P	P	X	P
8	Z M Quraishi	Zaheer	Male	Jan 5 200	P	P	P	P	P	P
8	Z M Quraishi	Faisal	Male	Feb 24 200	P	P	P	P	P	P
8	Z M Quraishi	Masood	Male	May 6 201	P	P	P	P	P	P
8	Z M Quraishi	Nuha	Female	Apr 11 201	P	X	X	P	X	P

Fig. 4.17 Reporting System

4.2 DATA BASE DESIGN

Table 4.1 DATA BASE DESIGN

Field Name	Data Type	Size	Allows Nulls	Constraints
Family No.	Int	5	No	No Keys
Head of Family	Char	20	Yes	No Keys
Child Name	Int	5	No	No Keys
Gender	Char	20	No	No Keys
Date Of Birth	Int	5	No	No Keys
Polio Given	Char	20	No	No Keys

4.3. CHOICE OF THE LANGUAGE

4.3. 1 Introduction to the J2ME Wireless Toolkit

The J2ME Wireless Toolkit supports the development of JAVA applications that run on the devices compliant with the Mobile Information Device Profile (MIDP), such as Cellular Phones, pagers and palmtops.

4.3. 2 Overview

The J2ME Wireless Toolkit supports a number of ways to develop MIDP applications. You can carry out the development process by running the tools from the command line or by using development environments that automate a large pan of this process.

The K Tool Bar, included with the J2ME Wireless toolkit, is a minimal development environment with a GUI for compiling, packaging, and executing MIDP applications. The only other tools you need are a third-party editor for your Java source files and a debugger.

An IDE compatible with the J2ME Wireless Toolkit provides even more convenience. For Example, when you use the Sun ONE Studio 4, Mobile Edition you can edit, compile, package, and execute or debug MIDP applications, all within the same environment. A list of IDEs those are compatible with the Wireless Toolkit. This section describes the phases of MIDP application development outside of editing, and how the toolkit contributes to these phases. The phases are illustrated in the following diagram:

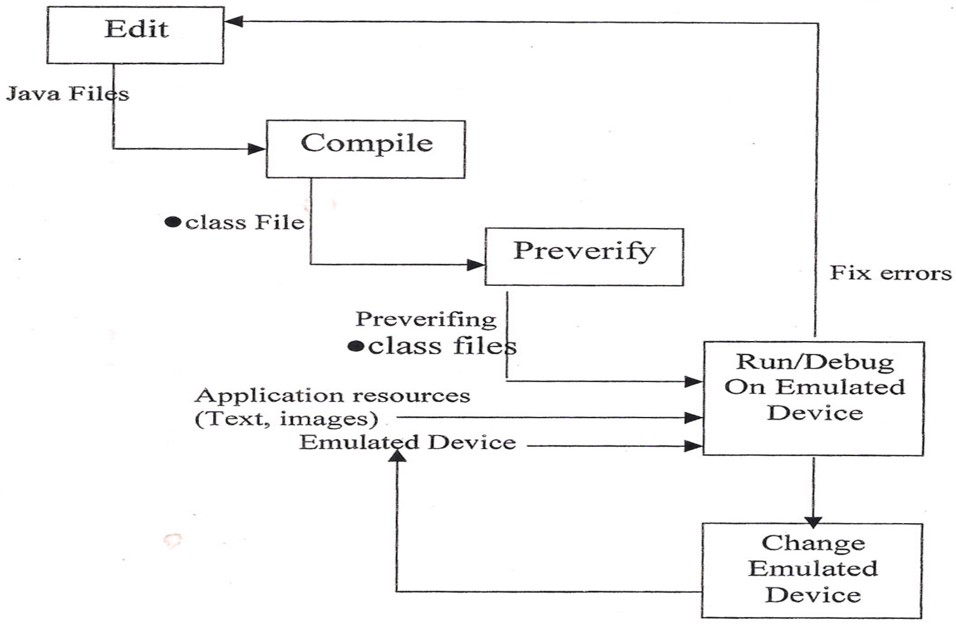


Fig. 4.18 Developing and Testing Application

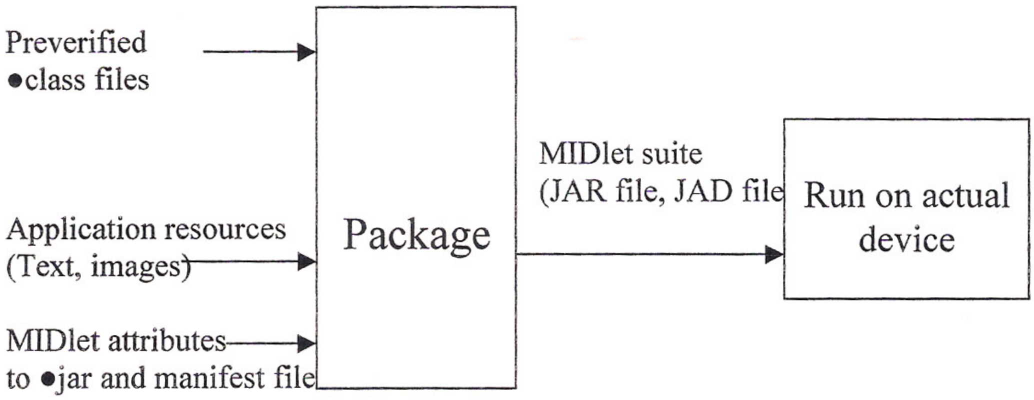


Fig. 4.19 Packaging an Application

4.3.3 Creating a New Project

To create a new project:

1. Choose File -> New Project from the menu or click **New** Project on the

The New Project dialog appears.

2. Type the name of the project in the Project Name Field, and the name of the main MIDlet class in the MIDlet Class Name field.

For Example, you might call the project newproject, and the MIDlet class might by myTestHelloc

3. Click Create Project.

The main window's title changes to include the name of the new project. The console indicates where to place your source, resource, and library files. The locations are consistent with the project file organization outlined in the above table.

4.3.4 Compiling and Revivifying

To compile and per-versify your source code:

Choose Project->Build or Click Build on the toolbar.

The sources are compiled against the MIDP and CLDC APIs, as well as any libraries in the project's lib\ folder and the toolkit's \apps\lib\folder.

4.3.5 Running

- Choose Project -> Run or Click Run on the toolbar
- The Emulator appears, running your MIDlet suite.
- The console displays system and trace output as a MIDlet suite executes.

4.3.6 To clear the console

Choose Edit -> Clear Console from the menu or click Clear Console on the toolbar.

4.3.7 Working with the Emulator

The Emulator shows, on your computer, how your MIDP applications operate on a variety of mobile devices. Consequently, you can test your application using the same platform you use to develop them, and defer testing on real devices until later in the development process.

The Emulator supports testing with several key features:

4.3.7.1 A variety of Devices

The Emulator can simulate several devices that take on a variety of form factors: cell phones, pagers, and palmtops. This helps you learn what type of experience users can expect and verify the portability of your application across different devices.

Tracing and debugging capabilities:

The emulator supports run-time logging of various events, such as garbage collection, methods call, and exceptions. You can also perform source-level debugging with an IDE while an application runs in the emulator.

4.3.7.2 Performance tuning:

The Emulator enables you to collect information that you can examine to optimize the performance of your MIDP application. Performance tuning utilities include profiling methods and monitoring memory usage and network traffic. Also include is the ability to adjust the speed settings for graphic rendering as well as VM speed emulation and network throughout.

It must be emphasized that having an Emulator does not completely free you from testing on your target devices. The Emulator does have some limitations.

4.3.7.3 Accuracy of Emulation:

The Emulator can only approximate a device's user interface, functionality, and good performance. For example, the Emulator cannot simulate processing speed, so an application may run faster or slower on a target device than it does on the Emulator.

4.3.7.4 Application Management:

The Emulator includes a sample implementation of an application manager, which supports the installation, maintenance, and removal of applications on a device.

However, this sample implementation does not emulate any application manager in particular, as the behavior of the application managers varies from device to device.

4.3.7.5 Example Devices:

The J2ME Wireless Toolkit includes emulations of various example devices to help you test the portability of your application. The examples possess a range of displays and features found in mobile information devices, and they all support the MIDP specification.

Tag	Description
DefaultColorPhone	Generic telephone with a color display
DefaultGrayPhone	Generic telephone with a gray-scale display.
MimmumPhone	Generic telephone with minimum display capabilities.
RIMJavaHandheld	RIM device from Research In Motion Ltd.
MotorolaJ85s	Motorola i85s telephone from Motorola, Inc.
Palms_pevice	Palm OS personal digital assistant from Palm, Inc.

CHAPTER 5

IMPLEMENTATION AND TESTING

5.1 DEVICES

The J2ME wireless Toolkit includes emulations of various examples devices to help you test the probability of your application. The examples possess a range of displays and features found in mobile information devices, and they all support the MIDP specification.

Tag: DefaultColorPhone, DefaultGrayPhone, MiimumPhone, RIMJava Handheld, Motorola_i85s, PalmOS_Devices.

5.2 CODING

The language Java 2 Platform Micro Edition (J2ME) provides easier way for coding than any other language. The intelligence facility coupled with J2ME (Sun ONE Studio 4, Mobile Edition) tools greatly reduces the amount of code the developer has to write.

5.3 TESTING

Testing is the phase where the errors remaining from the earlier phases must be detected. Hence, testing performs a very critical role for quality assurance and for assuring the reliability of software. There are different levels of testing, which performs different tasks and aim to test different aspects and acceptance testing.

5.3.1 Unit Testing

Unit testing focuses verification effort on smallest unit of software design -the module. The checklist for interface tests:

1. Number of input parameters equal to number of arguments
2. Parameter and argument attributes match?
3. Parameter and argument units system match?
4. Number attributes and order of arguments to built-in functions correct?
5. Any reference to parameters not associated with current point of entry?

When a module performs external I/O, additional interface tests must be conducted:

- File attributes correct?
- OPEN/CLOSE statements correct?

- Format specification matches I/O statement?
- Buffer size matches read size?
- End 0 of-file conditions handled?
- I/O errors handled?

The local data structure for module is a common source of errors. Test cases should be designed to uncover errors in the following;

1. Improper or inconsistent typing.
2. Erroneous initialization or default values.
3. Incorrect (misspelled or truncated) variables names.
4. Inconsistent data types.
5. Underflow, overflow, and addressing exceptions.

Unit testing is normally considered as an adjunct to the coding step. After the source level code has been developed, reviewed and verified for correct syntax, unit test case design begins.

5.3.2 Integration testing

The next level testing is often called integration testing. In this, many tested modules are combined into sub-system, which is then tested. The goal here is to see if the modules can be integrated properly, the emphasis being on testing interfaces between modules.

There may be question; if they all work individually, why it is doubtful that they will work when they put together?

The problem is "putting them together-interfacing. Data can be lost across an interface, one module can have an inadvertent, adverse affect on another, individually acceptable imprecision may be magnified to unacceptable levels, global data structures can present problems, and the list goes on and on.

5.3.3 System Testing

Here the entire software system is tested. The reference for this process is the requirements documents, and the goal is to see if the software meets its requirements.

5.3.4 Acceptance Testing

It is performed with the client demonstrate that the software is working satisfactorily.

5.4 SOFTWARE TEST PLANE

5.4.1 Objectives

The main objective of this test plan is to test the software to check that it fulfills all its requirements as specified.

5.4.2 Scope:

The scope of test plan is throughout the life-cycle of the product. Any changes made in the requirement leads to subsequent changes in the test plan.

5.4.3Test Deliverable

The output from the test plan would be the following

1. Test Cases for unit testing;
2. Test Log;
3. Test Status Report; and
4. Defect/Correction Report

The test status report includes the test conditions, test results, comments, and the defect ID. The defect report will be made for each defect and the developer has to make the correction as per the correction report. The defect report will contain the defect ID, the test item, the expected result, actual result, discrepancies and the correction report will also have the above data and the correction to be made in the respective file names.

5.5 IMPLEMENTATION

This is the most important phase of the System Development Life Cycle, where the actual implementation of the designed system occurs. Implementation is a process that includes all those activities that take place to convert an old system to a new system. There are three types of implementation:

1. Implementation of computer system to replace a manual system: the problems encountered are converting files, training users, creating accurate files, and verifying printouts for integrity.
2. Implementation of new computer system to replace an existing one: This is usually difficult conversion. If not properly planned there can be many problems. Some large computer systems have taken as long as a year to convert.
3. Implementation of modified application to replace an existing one: using the same computer. This type of conversion is relatively easy to handle provided there are no major changes in the files.

CHAPTER 6

CONCLUSION & FUTURE SCOPE

6.1 SCOPE AND ADVANTAGE OF THE SYSTEM

- Reaching the high risk groups and Vulnerable Communities
- Tracking missed children for Immunization
- Building Health system's capacities to deliver immunization program
- Developing Systems for community demand generation.
- Strengthening National and Regional Immunization Systems
- Instant Data Collection, Mapping & Digital Advisory Service Delivery
- Supported Timely Documentation and Use of Information

6.2 LIMITATIONS

The data and analysis have several limitations. First, we were unable to carry out an experimental design. Program/intervention areas (CMC) and comparison areas (non-CMC) were not selected at random. CMC areas are a group of villages within a block that have been purposively selected because they have the most difficulty achieving vaccination program targets. They tend to have a much higher proportion of families that are resistant to allowing their children to be vaccinated with polio vaccine. Therefore, it would be understandable if values of the three performance indicators selected would be lower, on average, in CMC areas compared to non-CMC areas even after considerable social mobilization effort. However, if performance in CMC areas approaches or surpasses the performance in non-CMC areas, we might judge the social mobilization efforts as having had positive effects.

Another limitation of the data is the difficulty of analyzing trends. We expect performance to vary up and down by season. For example, during harvest seasons we might find fewer household members and eligible children at home; the number of vaccinations would decrease in this situation while the denominator would remain high. In addition, the denominator (number of eligible children under five years of age to be vaccinated) changes from SIA to SIA. The denominator is the number of children under five vaccinated during the prior campaign. We assume this number to be a more accurate estimate of the actual number of children under five living in the area than the census because health workers going house-to-house on a regular basis to count this number of children obtain this number. In this situation, we would expect vaccination performance to move up and down from SIA to SIA as an artifact of the changing denominator. For this reason, the analysis looked at the mean difference in performance between CMC and non-CMC areas rather than comparing trends between the two comparison groups.

There is limitation in assessing the effects of social mobilization on performance of supplementary immunization activities such as national or sub-national immunization days. SIAs, while necessary, are not sufficient. Many other factors affect progress of the polio eradication effort such as routine immunization efforts, sanitation, and vaccine efficacy in crowded, unsanitary areas.

There is no doubt that Embedded Games and application for Mobile Devices are very useful. But, it has some limitations. Some of the limitations of Embedded Games and application for Mobile Devices are:

- Virtual Machine should be installed.
- Multiple users cannot play the same game at the same time in same device.
- After certain time period game will be terminate (game over).

6.3 CONCLUSIONS

The SM Net was established in the northern Indian state of Uttar Pradesh---in high-risk villages and urban areas in high transmission areas of a country endemic for polio---for the purpose of motivating families to have all their children under five years of age receive polio vaccine during routine immunization services and during supplemental immunization activities (SIAs). Even though we expect it to be more difficult to vaccinate in SM Net villages as compared to non-SM Net villages, performance was as good and often better in CGPP programs areas vs. non-CGPP program areas. This was observed even though villages in CGPP program areas were purposively selected because they had a high proportion of families (relative to villages in non-CGPP program areas) that were resistant to having their children vaccinated against polio. The performance of the CGPP Project on indicators assessed appears to validate the social mobilization efforts of the SM Net in India toward eradication of polio. The CGPP model---achieving scale through a partnership of PVOs and NGOs under the leadership of an independent Secretariat---appears promising and should be explored further as a model for social mobilization and other health programming efforts. Additional analysis as to what particular social mobilization efforts are associated with better performance on the vaccination indicators included in this analysis is recommended.

While not conclusive, the results suggest that the social mobilization efforts of the SM Net and the CORE Group are helping to increase vaccination levels in high-risk areas of Uttar Pradesh. Vaccination outcomes in CORE Group areas were equal or higher than in non-CORE, non-SM Net areas. This occurred even though SM Net areas are those with more community resistance to polio vaccination and/or are have harder-to-reach populations than non-SM Net areas. Other likely explanations for the relatively good vaccination performance in SM Net areas are not apparent.

Embedded Games and application for Mobile Devices has unique application and advantages. It provides a very advance and amusing games, which are more attractive and full of entertainment and provides relaxation of the player of the game.

6.4 FUTURE SCOPE

- ❑ Integrator with National Health

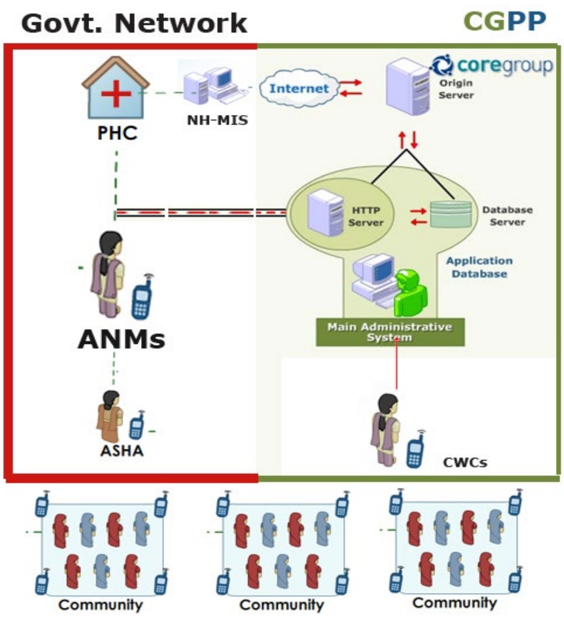


Fig 6.1 Integrator with National Health

- ❑ GPS Mapping of Villages for Surveillance for High Risk, other Behavioral Patterns and GIS Data Analysis

- ❑ Mobile Training Kit for CMCs

Fig 6.2 Mobile Training Kit for CMCs

- ❑ Mobile App Center for Community Programs like Mothers Meetings , general awareness programs.
- ❑ Scaling the Project for Universal Immunization Globally
- ❑ Developing Community level Digital Tools – for SMS, Feature phone and Smartphone based
- ❑ Gaming with Live Data for Training and Simulations

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